

**M. TECH.
(SEM I) THEORY EXAMINATION 2022-23
ADVANCED ALGORITHMS**

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 7 = 14**

- (a) What is asymptotic notation? Explain Big Oh notation?
- (b) What is recurrence relation? How a recurrence is solved using master's theorem?
- (c) Write down the properties of Red Black tree.
- (d) Write down design issue of parallel algorithm?
- (e) Differentiate Backtracking algorithm with branch and bound algorithm.
- (f) Discuss Greedy approach with suitable example.
- (g) Write down the characteristics of B tree.

SECTION B**2. Attempt any three of the following:****7 x 3 = 21**

- (a) Solve the following recurrence relation
 - (i) $T(n) = 4T(n/2) + n^2$
 - (ii) $T(n) = 2T(\sqrt{n}) + \log n$
- (b) Sort the following list of elements in increasing order, using merge sort technique and also analysis the time complexity of merge sort algorithm
14, 33, 21, 45, 67, 20, 40, 59, 12, 36.
- (c) Discuss the various cases for insertion of key in red-black tree for given sequence of key in an empty red-black tree- **5, 16, 22, 25, 2, 10, 18, 30, 50, 12, 1**
- (d) Discuss merging on EREW model with suitable example.
- (e) Explain Depth First search algorithm with suitable example.

SECTION C**3. Attempt any one part of the following:****7 x 1 = 7**

- (a) Write down the algorithm for Max Heapify in heap sort .Write its time complexity also.
- (b) Describe the properties and requirements of a good algorithm.

4. Attempt any *one* part of the following:

7 x 1 = 7

- (a) What do you mean by parallel sorting networks? Also discuss the enumeration sort algorithm.
- (b) Explain Branch & Bound method. How 0/1 Knapsack problem can be solved using branch and bound method?

5. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Find the optimal solution to the Knapsack instances $n = 5$, $w = (5, 10, 20, 30, 40)$ $v = (30, 20, 100, 90, 160)$ and $W = 60$, by using Greedy approach (Fractional Knapsack)
- (b) What is sum of subset problem? Draw a state space tree for Sum of subset problem using backtracking? Let $n=6$, $m=30$ and $w [1:6] = \{5, 10, 12, 13, 15, 18\}$

6. Attempt any *one* part of the following:

7 x 1 = 7

- (a) What are the factors performance measures of Parallel Algorithm?
- (b) What is parallel searching and also explain the CREW searching?

7. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Describe Breadth First search algorithm with suitable example.
- (b) Explain P, NP, NP Complete and NP hard problems with suitable examples.